

Aula 5

Monday, March 9, 2026 3:58 PM

Arithmetic operation

↳ Operações entre imagens

$$- s(x, y) = f(x, y) + g(x, y)$$

$$- d(x, y) = f(x, y) - g(x, y)$$

$$- p(x, y) = f(x, y) \cdot g(x, y)$$

$$- v(x, y) = f(x, y) \div g(x, y)$$

↳ Operações pixel a pixel

Operações sobre o brilho

$$\left[255 \cdot \frac{x - \min}{\max - \min} \right] \rightarrow \text{transformação linear}$$

$\eta(x, y)$ = ruído gaussiano

(média 0 e desvio padrão 1)

$$g(x, y) = f(x, y) + \eta(x, y)$$

$$\frac{1}{N} \sum_{i=1}^N \eta_i(x, y) \xrightarrow{N \rightarrow \infty} 0$$

Operador T

$$\boxed{y} \leftarrow T(\boxed{x})$$

$$T: f \rightarrow \tilde{f}$$

↳ conj. de todas as possíveis imagens

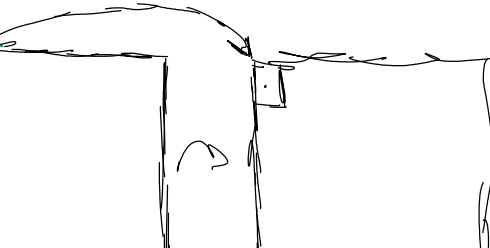
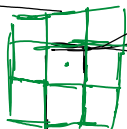
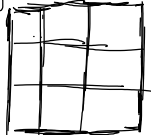
T: pode ser linear ou não

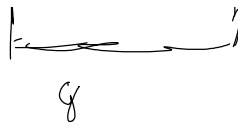
def: T é um operador linear se

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

Alguns possíveis Ts

imagem 3x3





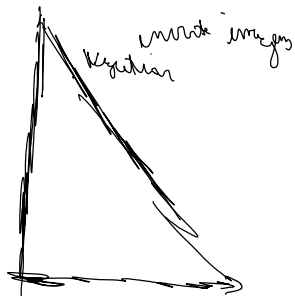
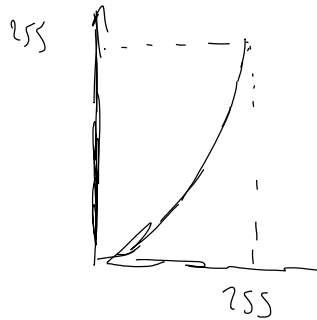
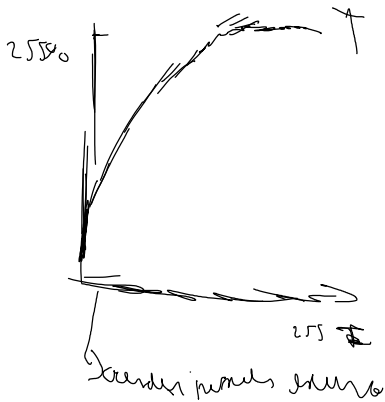
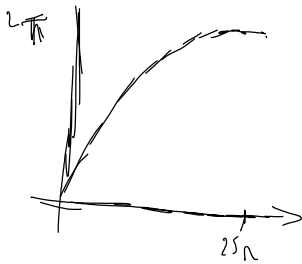
Dependem da vizinhança

$$h(x, y) = \begin{cases} 1 & \text{se } f(x, y) > 0 \\ 0 & \text{caso contrário} \end{cases}$$

$$h(x, y) = T_\theta(f(x, y))$$

não depende da vizinhança

$$s \leftarrow T(n)$$

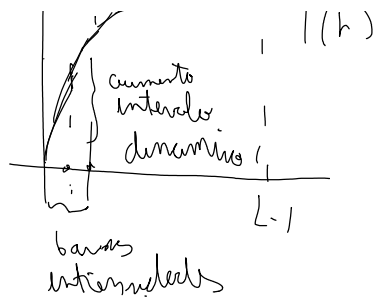


Cap 3

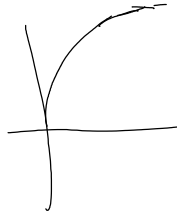
Power Law (gamma) Transformation

$$s = cr^{\gamma}$$

L-1 +



Gamma transf.
 $b_5 = c \cdot h^f$
 $0 < f < 1$



$f > 1$

